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Simulation and Estimation in a Markov Model of Breast Cancer

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GENAI DECLARATION

We have used artificial intelligence tools from OpenAI, Anthropic, and Google for synthesizing, coding and to assist with writing. In accordance with DTU's guidelines for documenting the use of Generative AI, we acknowledge this usage.

Welcome to our submission of Project 1 for the course Stochastic Simulation at DTU. All simulations were implemented in Python and use a fixed random seed of 67, so the results in this report are exactly reproducible.

PART 1: A DISCRETE-TIME MODEL

After breast cancer surgery, a woman's health each month is modeled as one of 5 states: cancer-free (1), local recurrence (2), distant metastasis (3), both (4), or death (5). Each month she either stays in her current state or moves to a higher one, with the probabilities shown in Figure 1 (self-loops, i.e. staying put, are omitted). We call the number of months until a woman reaches state 5 her *lifetime*, T .

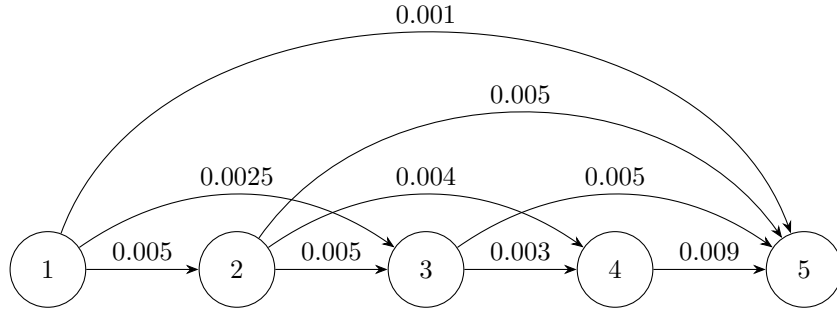


Fig. 1: Monthly transition probabilities between the 5 states.

Task 1

We simulated 1000 women, all starting in state 1, and ran each one forward month by month until she reached state 5. The lifetimes T are shown in Figure 2.

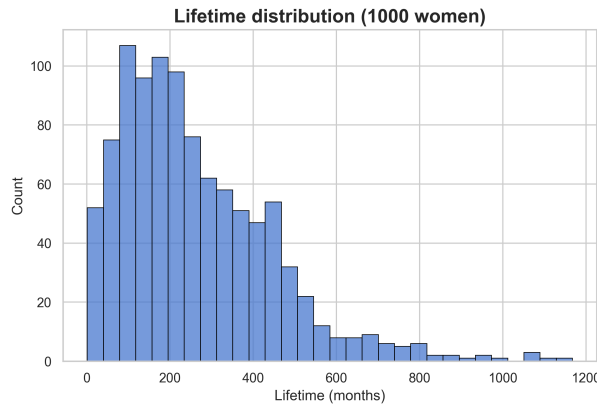


Fig. 2: Lifetime distribution of 1000 simulated women.

The average lifetime was about 262.8 months (roughly 22 years). A lot of women die between 100 and 300 months, but a few live much longer, giving the histogram a long tail to the right.

We also checked how often the cancer comes back locally. In 57% of women, the cancer reappeared at the original surgery site at some point. If we also count women who reached the "both" state without ever being recorded as local-only (i.e. local and distant metastasis appeared at the same time), this number rises to 68%.

Task 2

As a check on the simulation, we looked at the distribution over the 5 states at a single point in time, $t = 120$ months, and compared it to what the model predicts theoretically. For each of the 1000 women we recorded which state she was in at month 120 (a woman who died earlier than this is counted as being in the death state). The theoretical distribution at the same time point is $p_{120} = p_0 P^{120}$, where $p_0 = (1, 0, 0, 0, 0)$ since every woman starts in state 1.

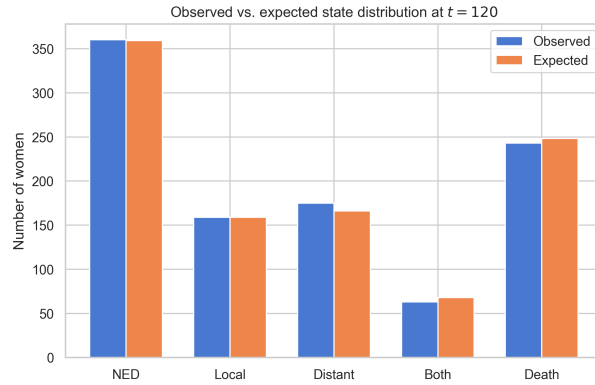


Fig. 3: Observed vs. expected number of women in each state at $t = 120$.

The observed and expected counts, shown in Figure 3, are close across all 5 states. To check this formally, we ran a chi-square goodness-of-fit test comparing the observed counts to the expected counts. This gave $\chi^2 = 0.92$ with 4 degrees of freedom, and a p-value of 0.92. Since this is far from significant, we have no evidence that the simulated distribution at $t = 120$ deviates from the theoretical one, which is what we would expect if the simulation correctly implements P .

Task 3

Task 2 only checked the simulation at a single point in time. Here we check the full lifetime distribution instead. The lifetime T should follow a discrete phase-type distribution, which also gives us a theoretical mean lifetime of 262.4 months. This matches the simulated mean of 262.8 months closely.

As we learned in the lecture, a chi-square test needs an expected count of at least 5 in every bin to be reliable. Since a lifetime can take on hundreds of different values, binning by exact month would leave most bins with an expected count close to 0. To avoid this, we instead split the theoretical distribution into 10 equally likely bins (deciles), so each bin is expected to contain about 100 of the 1000 women regardless of how spread out the distribution is. We then compared the actual number of women in each bin to this expected number, using a chi-square goodness-of-fit test.

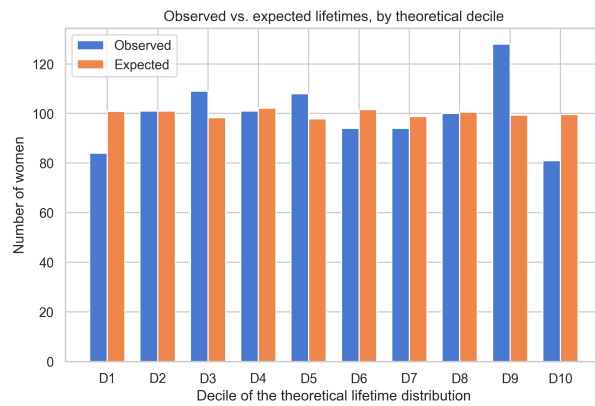


Fig. 4: Observed vs. expected number of women in each decile of the theoretical lifetime distribution.

This gave $\chi^2 = 17.65$ with 9 degrees of freedom and a p-value of 0.039. On its own, this looks bad: a p-value below 0.05 would normally mean we reject the idea that the simulation matches the theoretical distribution.

Before concluding that, we reran the same test with 20 different random seeds. The p-values varied a lot, from about 0.09 to 0.79, with most of them well above 0.05. This is exactly what we would expect even if the simulation is correct: by chance, about 1 in 20 seeds will give a p-value below 0.05, and the seed used for this report just happened to be one of the unlucky ones.

A single “significant” result like that doesn’t mean much on its own, especially when we already have strong reason to trust the simulation, since it’s just sampling directly from P . Much like the statistician pictured here, we’re not about to abandon a well-supported belief over one unlucky roll of the dice.

So we conclude that the simulated lifetimes do follow the theoretical distribution. The one low p-value is normal sampling variation, not a sign that the simulation is wrong.

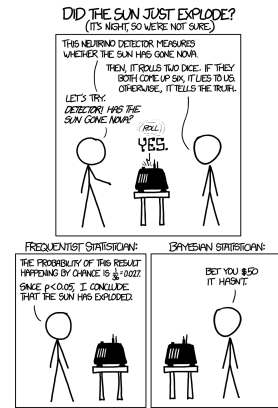


Fig. 5: xkcd #1132, (Randall Munroe).

Task 4

Here we estimate the expected lifetime of a woman who is still alive at month 12, and whose cancer has already reappeared (locally, distantly, or both) by that point. Since a woman’s state can never move back to NED (“no evidence of disease”, i.e. still cancer-free) once she has left it, this is the same as asking: what is her state at exactly month 12? If it is local recurrence, distant metastasis, or both, she qualifies; if it is still NED or already death, she does not.

We used rejection sampling to estimate this. We simulated women one at a time, exactly as in Task 1, but this time we only kept a woman if her state at month 12 met the condition above. Women who did not meet it were discarded. We repeated this until we had collected 1000 women who qualified, and then took the average of their total lifetimes (i.e. the full time until death, not just the first 12 months).

Out of 11 907 simulated women, 1000 met the condition, an acceptance rate of about 8.4%. This confirms that this is a fairly rare group: the vast majority of women are either still cancer-free or are already dead by month 12.

The estimated expected lifetime for this group was about 172.5 months, with a 95% confidence interval of (162.8, 182.1).

This number is interesting to compare to the overall average lifetime from Task 1, which was about 262.8 months: the conditional lifetime here is much lower. This makes sense, since we are explicitly excluding the women who stay in NED, and those are exactly the women who tend to live the longest, as they avoid the higher death rates that come with recurrence. Looking at P , the death probability per month goes from 0.001 in NED, up to 0.005 in local recurrence or distant metastasis, and 0.009 once both are present. So by requiring the cancer to have already reappeared by month 12, we are left with only the women whose disease is progressing more aggressively, which pulls the expected lifetime down compared to the full population in Task 1.

Task 5

Here we want to know what fraction of women die within the first 350 months, $p = P(T \leq 350)$. We first set up the two estimators we want to compare, and then look at how well they actually did.

We simulate women one at a time using the same per-woman simulation as in Task 1 (starting in state 1 and stepping forward month by month until death), but now in batches of 200 women rather than 1000. For each woman we look at $X_i = 1$ if she died by month 350, and $X_i = 0$ otherwise. The plain estimate of p is then just the average of these 200 values, $\bar{X}$. We do this for 100 separate batches of 200 women, giving us 100 estimates of p , so we can see how much these estimates vary.

To get a more precise estimate, we use a control variate: a quantity that is correlated with what we’re estimating, but whose true average we already know. A long-living batch of women will tend to have fewer deaths by month 350, so a natural choice is each woman’s full lifetime, Z_i . We already know its

true mean from Task 3, $\mu_Z = 262.4$ months. The control-variate estimator adjusts the plain estimate using this extra information:

$$Y_i = X_i + c(Z_i - \mu_Z), \quad c = -\frac{\widehat{\text{Cov}}(X_i, Z_i)}{\widehat{\text{Var}}(Z_i)},$$

where $\widehat{\text{Cov}}$ and $\widehat{\text{Var}}$ are estimated from the same 200 women. If a batch happens to live longer than average (Z_i above μ_Z), this correction nudges the death-fraction estimate back down (since c is negative), and vice versa. Averaging Y_i over the 200 women gives the control-variate estimate of p , and we again repeat this for 100 batches.

With both estimators set up, we can now compare how they actually performed. The plain estimator gave a mean of 0.7357 with 95% CI (0.7289, 0.7425) and variance 1.213×10^{-3} . The control-variate estimator gave a mean of 0.7393 with 95% CI (0.7356, 0.7431) and variance 3.63×10^{-4} . Both agree on roughly the same value of p , which is what we'd expect since both estimators are unbiased, but the control-variate confidence interval is noticeably tighter.

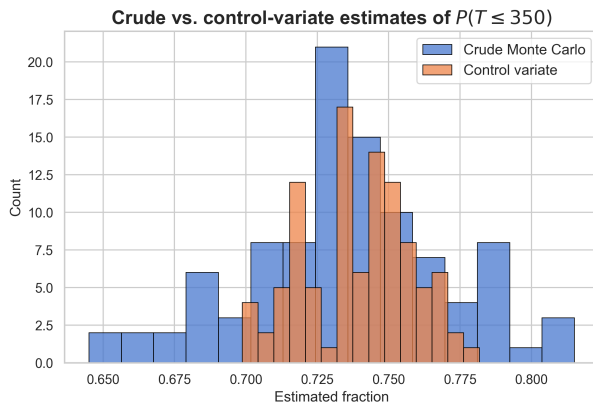


Fig. 6: Distribution of the 100 plain and control-variate estimates of $P(T \leq 350)$.

Overall, using the lifetime as a control variate reduced the variance of the estimator by about 70%, a factor of roughly 3.3. This makes sense: lifetime and death-by-350 are strongly correlated, so correcting for how long a particular batch of women happened to live removes a large part of the random noise in the estimate, at no extra simulation cost.

Task 6

Our simulation in Tasks 1–5 assumes the women’s health behaves like a Markov chain with one fixed matrix P used forever. Hidden in that are several separate assumptions. First, the Markov property: what happens next depends only on which state she’s in right now, not on how she got there. Second, time-homogeneity: P never changes over time, the chance of dying from local recurrence is the same in month 1 as in month 300. Together these mean how long she’s already spent in a state tells us nothing about when she’ll leave it. On top of that, her status is only checked once a month rather than continuously, and her health is fully summarized by one of 5 fixed categories rather than anything more detailed. The chain is also one-way: death can’t be left and recurrence can’t be undone, so there’s no path back to NED. Finally, every woman is assumed to behave identically, regardless of age, tumor type, or treatment, and one woman’s outcome is assumed to tell us nothing about another’s.

Some of these hold up better than others. Time-homogeneity is one of the weaker ones: a quick google search reveals that real recurrence hazard peaks at around 15% per year in the first 1–2 years after surgery and then declines steadily for years afterward, nowhere close to the constant monthly risk our model assumes. The Markov property has a similar problem, since real progression probably depends on more than just the current state. Using the same matrix for every woman is likely the least realistic assumption of all: a google search reveals that tumor grade is considered the single most influential factor

in predicting recurrence, with age and treatment type also playing a role. Lumping everyone into one P throws all of that information away. On the other hand, the one-way absorbing structure is fairly realistic, since patients generally aren't reclassified back to cancer-free once metastasis is confirmed. Independence between women is also a decent approximation, unless they happened to share doctors or hospitals or were genetically related. The monthly time step and the 5-state simplification are less about realism and more deliberate simplifications for tractability.

These assumptions could be relaxed, generally at the cost of losing the closed-form formulas we used in Tasks 2 and 3 and needing a more complicated model. Time-homogeneity could be dropped by letting P depend on time since surgery, or by moving to a semi-Markov model where time spent in each state follows something other than a geometric distribution, the same idea behind the Erlang-distributed sojourn times mentioned later in this project (Task 11), just generalized to discrete time. The monthly time step could be replaced with continuous time, which is exactly what Part 2 of this project does. Using the same matrix for everyone could be fixed by giving each woman her own transition rates based on her age, tumor stage, and treatment. We looked up the original paper this project is based on (Putter et al., 2006), and that's exactly what they did with real patient data, at the cost of needing a regression-style model per transition and much more data than our simple simulation uses. And if we wanted to drop the Markov property itself rather than patch it, we could replace the matrix with a sequence model like an LSTM or transformer trained on a woman's full visit history, similar to what some recent papers do for breast cancer prognosis from electronic health records, though that only makes sense once fitting to real data rather than running a closed-form simulation like ours, and gives up all of the interpretable, closed-form structure we relied on earlier.

PART 2: A CONTINUOUS-TIME MODEL

Task 7

For the second part of the project, we switched to the continuous-time model and once again simulated a 1000 women, all starting in state 1. Each woman was run forward in continuous time, waiting an exponentially distributed amount of time in each state before jumping, until she reached state 5. The lifetimes T are shown in Figure 7.

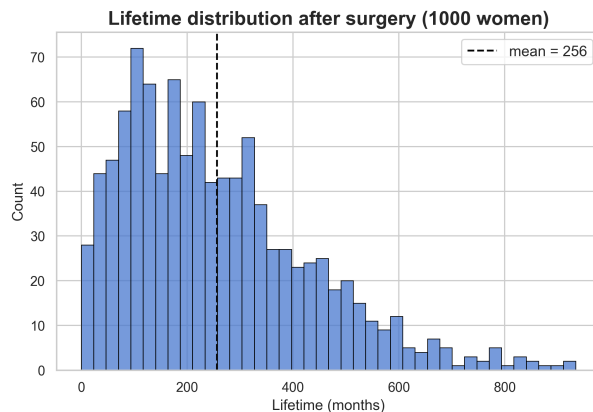


Fig. 7: Lifetime distribution of 1000 simulated women (continuous-time model).

The mean lifetime and its spread are summarised in Table I. The lifetimes are widely spread, with a long right tail where a few women live much longer than the rest. The proportion of women whose cancer had reappeared distantly within the first 30.5 months after surgery was found to be about 9%, counting those who had entered the distant metastasis state or the "both" state.

Task 8

We compared the empirical lifetime distribution function from our 1000 simulations to the theoretical continuous-time phase-type distribution function, $F_T(t) = 1 - p_0 \exp(Q_s t) \mathbf{1}$. The two are plotted together in Figure 8.

TABLE I: Lifetime after surgery for 1000 simulated women (continuous-time model). The mean interval is a t -interval, the standard-deviation interval a bootstrap interval.

Quantity	Estimate	95% CI
Mean lifetime (months)	256	[245, 267]
Std. deviation (months)	176	[167, 185]

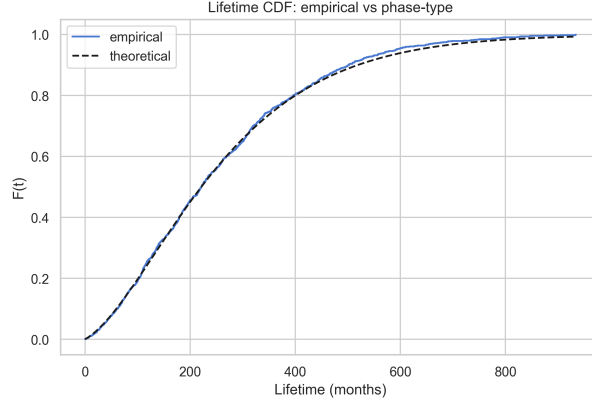


Fig. 8: Empirical (step) and theoretical (dashed) lifetime distribution functions for the continuous-time model.

Since the lifetimes are continuous, the empirical and theoretical distribution functions were compared using a Kolmogorov–Smirnov test, reported in Table II. The test statistic is small and the p -value is far above 0.05, so we cannot reject the hypothesis that the simulated lifetimes follow the theoretical phase-type distribution. This agrees with the close overlap of the two curves in Figure 8.

TABLE II: Kolmogorov–Smirnov test of the empirical lifetimes against the theoretical phase-type distribution function.

Comparison	KS stat	p -value
Empirical vs. theoretical CDF	0.017	0.930

Task 9

We simulated 1000 women who received the preventive treatment and estimated their survival function with the Kaplan–Meier estimator, $\hat{S}(t) = (N - d(t))/N$. Figure 9 shows this curve together with the Kaplan–Meier estimate for the 1000 untreated women.

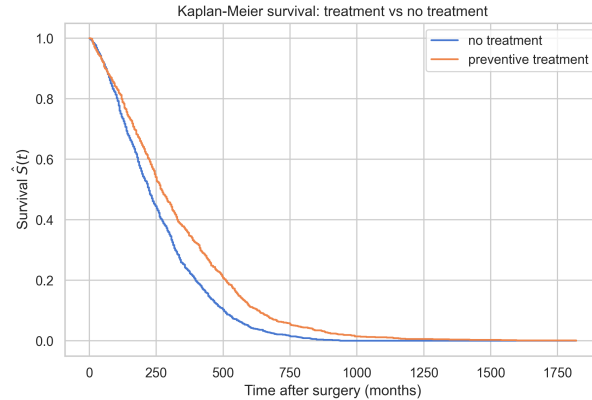


Fig. 9: Kaplan–Meier survival functions for women with and without the preventive treatment.

The treatment does appear to have an effect. Across the whole follow-up period the survival curve for the treated women lies above the curve for the untreated women in Figure 9, so at any given time a larger fraction of treated women are still alive. In other words, the treatment shifts the lifetime distribution towards longer survival.

Task 10

The preventive treatment was found to have a significant effect on the survival function. We tested the two survival functions for equality with a log-rank test, reported in Table III. The test statistic is very large and the p -value is far below 0.05, so we reject the null hypothesis that treated and untreated women have the same survival function. This confirms the visual difference between the two Kaplan–Meier curves in Task 9.

TABLE III: Log-rank test comparing the survival functions of treated and untreated women (1 degree of freedom).

Test	χ^2	p -value
Log-rank (treated vs. untreated)	57.5	3.4×10^{-14}

Task 11

The assumption eliminated by going from the discrete- to the continuous-time model is the fixed monthly time grid. In the discrete model a woman can only change state at integer months, which implicitly forces the time spent in a state to follow a geometric distribution. The continuous-time model removes this artificial discretisation: transitions may happen at any instant, so we no longer pretend that nothing happens between monthly steps.

The assumption added in return is that the sojourn time in each state is exponentially distributed, with rate $-q_{ii}$. This is the memoryless property: the rate of leaving a state is constant over time and does not depend on how long the woman has already spent there. For many diseases this is unrealistic, since the risk of progression usually changes with the time since surgery.

The model can be extended to give Erlang-distributed sojourn times by replacing a single state with a chain of k sub-states, each left at the same rate λ . The time to pass through all k sub-states is then a sum of k independent exponentials, i.e. an Erlang(k, λ) distribution, whose hazard increases with time. This is a phase-type expansion: the model remains a continuous-time Markov chain, so all of our methods still apply, at the cost of extra latent states.

PART 2: ESTIMATION

Task 12

We simulated 1000 women in the continuous-time model (the same Q as in Part 2) until death and recorded each woman’s state every 48 months, giving a time series $Y^{(i)} = (X(0), X(48), \dots, 5)$ that begins in state 1 and ends in death, for example $(1, 1, 1, 2, 5)$. These 1000 series are treated as the only available observations for the rest of Part 3.

The length of each observed series (the number of screenings recorded for a given woman) ranged from 2 to 21, with a mean of about 7. Its distribution is shown in Figure 10, where each bar counts the women whose record has a given length.

Because every series ends in death, this length is itself a coarse measure of survival. A record of length L means the woman was last seen alive at the $48(L - 2)$ -month screening and recorded dead at the $48(L - 1)$ -month one. Each additional unit of length therefore corresponds to surviving roughly one more 48-month interval, so the bars are simply survival times binned at a 48-month resolution. This is why the distribution mirrors the lifetime distribution from Part 2: a pronounced peak at short records and a long right tail.

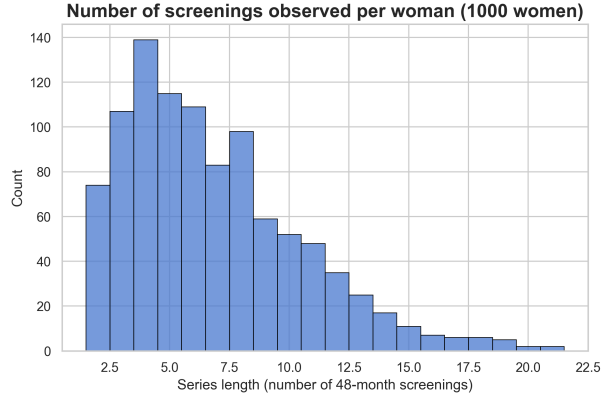


Fig. 10: Distribution of the observed-series length (number of 48-month screenings) across the 1000 simulated women. Each bar counts the women with that record length; since every series ends in death, the length is survival measured in 48-month screening intervals.

Task 13

From the coarse observations alone, the transition rates can be estimated with the unbiased estimator $q_{ij} = N_{ij}/S_i$, where N_{ij} is the total number of jumps from state i to j and S_i the total sojourn time in state i . Both quantities are hidden, since we only see the state every 48 months, so we recovered them with a Monte-Carlo EM algorithm. Starting from an initial guess $Q^{(0)}$ (a small rate on every transition the model permits), each iteration performs two steps. In the E-step, for every observed interval $y_k \rightarrow y_{k+1}$ we simulate a complete CTMC trajectory over 48 months under the current Q and reject it unless it ends in the observed state y_{k+1} ; the accepted bridges accumulate N_{ij} and S_i across all 1000 women. In the M-step we set $q_{ij} = N_{ij}/S_i$ and complete the diagonal with equation (1). As a convergence target we monitored the infinity-norm change between successive iterates, $\|Q^{(k)} - Q^{(k+1)}\|_\infty$, against the suggested threshold 10^{-3} .

The infinity-norm change drops sharply after the first iteration and then settles around a Monte-Carlo noise floor of about 4×10^{-4} , as shown in Figure 11. Because each E-step reconstructs only one trajectory per interval, the iterates never stop moving entirely: the change first falls below 10^{-3} already at iteration 2, but it does not keep decreasing, so this crossing reflects the size of the random step between two iterations rather than true convergence. Stopping at the first crossing would therefore report a single noisy iterate. Instead we fixed the run length at 30 iterations, well past the point where the change reaches the noise floor, discarded the first five iterates as burn-in, and took the estimate to be the average of the remaining iterates, which averages out the per-iteration Monte-Carlo noise.

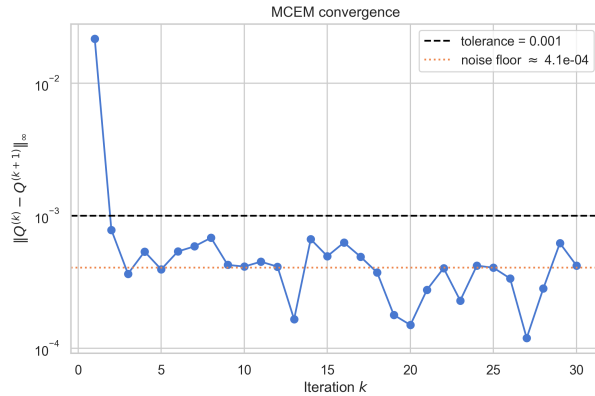


Fig. 11: Convergence of the MCEM algorithm: the infinity-norm change in Q between successive iterations (log scale) falls below the 10^{-3} tolerance after the first iteration and then fluctuates around a Monte-Carlo noise floor (dotted) rather than decreasing further.

The recovered rates reproduce the true Q closely, as summarised in Table IV, where the uncertainty is the standard deviation of the post-burn-in iterates. The remaining differences from the true values (at most about 10^{-3}) are larger than this Monte-Carlo spread and are consistent with the sampling error of estimating rates from only 1000 women, confirming that the original parameters can be approximately recreated from observations taken only every fourth year.

TABLE IV: True versus MCEM-recovered transition rates q_{ij} (months $^{-1}$) for the transitions the model allows. Uncertainty is the standard deviation of the post-burn-in iterates.

Transition $i \rightarrow j$	True q_{ij}	Recovered q_{ij}
$1 \rightarrow 2$	0.00500	0.00498 ± 0.00003
$1 \rightarrow 3$	0.00250	0.00248 ± 0.00010
$1 \rightarrow 5$	0.00100	0.00082 ± 0.00008
$2 \rightarrow 3$	0.00500	0.00479 ± 0.00026
$2 \rightarrow 4$	0.00400	0.00488 ± 0.00022
$2 \rightarrow 5$	0.00500	0.00554 ± 0.00025
$3 \rightarrow 4$	0.00300	0.00301 ± 0.00010
$3 \rightarrow 5$	0.00500	0.00525 ± 0.00014
$4 \rightarrow 5$	0.00900	0.01004 ± 0.00029